



HITEC UNIVERSITY TAXILA

AI LAB PROJECT REPORT

Cricket Match Outcome Prediction

Submitted by:

Muhammad Usman (24-cys-030)

Muhammad Faizan Ali (24-cys-006)

Submitted to: Sir Mubashir Iqbal

➤ Introduction:

1. Broad Domain Context: AI/ML in Sports Analytics:

Artificial Intelligence (AI) and Machine Learning (ML) have rapidly transformed the landscape of sports analytics, enabling data-driven decision-making across performance evaluation, fan engagement, injury prevention, and strategic planning. Modern sports generate massive volumes of structured and unstructured data, including player statistics, ball-by-ball match records, weather conditions, and contextual game variables. Leveraging this data through AI techniques allows stakeholders to uncover complex patterns that are often imperceptible to human analysts. The global sports analytics market has witnessed exponential growth, driven by the increasing adoption of predictive modeling, real-time data processing, and cloud-based analytics platforms. Industry reports estimate that the sports analytics market is projected to exceed **USD 20 billion by the end of this decade**, reflecting the growing reliance on intelligent systems for competitive advantage.

Within this ecosystem, cricket occupies a unique position. As one of the most widely followed sports globally particularly in South Asia, Australia, and the United Kingdom cricket generates exceptionally rich datasets due to its event-based structure and multiple game formats. Each match encapsulates hundreds of micro-events, including deliveries, scoring shots, dismissals, and fielding actions. AI and ML models have the potential to revolutionize cricket by providing accurate match outcome predictions, optimizing team selection, and enhancing tactical decisions. Consequently, cricket match outcome prediction has emerged as a compelling application domain for machine learning research, blending statistical rigor with real-world impact.

Cricket, as one of the most widely played and watched sports globally, represents a particularly compelling domain for AI-driven analytics. With billions of fans worldwide and professional leagues generating vast amounts of structured data, cricket provides an ideal environment for applying machine learning techniques. The sport's event-driven nature—characterized by ball-by-ball records, player-level statistics, and contextual match conditions—creates opportunities for advanced predictive modeling. Consequently, cricket match outcome prediction has become a growing area of research interest within sports analytics and machine learning communities.

2. Narrow Topic Focus: Cricket Match Outcome Prediction:

Despite the abundance of cricket data, predicting the outcome of a cricket match remains a challenging and practically significant problem. Cricket matches are influenced by a diverse set of factors, such as team composition, individual player form, pitch conditions, weather variability, venue characteristics, toss decisions, and match format (Test, ODI, or T20). For example, a single player's performance or an unexpected weather interruption can dramatically alter the match trajectory. Accurate match outcome prediction is therefore not only valuable for teams and coaches but also for broadcasters, analysts, fantasy sports platforms, and betting industries.

The importance of this sub-problem is underscored by its financial and strategic implications. Fantasy sports platforms and broadcasting agencies rely heavily on predictive insights to enhance user engagement and monetization. At the professional level, national teams invest significant resources in performance analytics to improve winning probabilities. For instance, in high-stakes tournaments such as the ICC Cricket World Cup or Indian Premier League (IPL), marginal predictive improvements can translate into substantial competitive and economic gains. Hence, developing reliable machine learning models for cricket match outcome prediction is both a technically challenging and economically relevant research problem.

From an economic perspective, the rise of fantasy sports platforms and online betting markets has further amplified the demand for reliable prediction systems. Millions of users rely on predictive insights to make informed decisions, and even marginal improvements in prediction accuracy can significantly impact user engagement and revenue. Given these practical implications, developing robust and accurate machine learning models for cricket match outcome prediction is both a technically challenging and socially relevant research problem.

3. Research Gap and Motivation:

Although several studies have explored cricket match prediction using machine learning, notable research gaps remain. Early works primarily employed traditional statistical methods and shallow machine learning algorithms such as Logistic Regression, Naïve Bayes, Support Vector Machines (SVM), and Decision Trees. While these models demonstrated moderate predictive accuracy, they often relied on limited feature sets and made simplifying assumptions about match dynamics. For example, some studies focused exclusively on historical win-loss records while ignoring contextual variables such as venue effects or recent player performance.

More recent studies have introduced ensemble models and deep learning techniques; however, many suffer from methodological limitations. Several authors explicitly acknowledge challenges such as **class imbalance**, **overfitting on small datasets**, and **poor generalization across formats and venues**. Additionally, a significant portion of existing research evaluates models using accuracy alone, neglecting more robust metrics such as F1-score or AUC, which are critical when match outcomes are unevenly distributed. Furthermore, many studies are constrained by outdated datasets, failing to incorporate recent matches, evolving team strategies, and modern playing styles.

The motivation for this research arises from these unresolved challenges and recent advancements in data availability and computational power. The emergence of comprehensive ball-by-ball datasets, improved feature engineering techniques, and powerful ensemble learning algorithms provides an opportunity to design more robust and generalizable prediction models. This research seeks to bridge the gap by addressing data imbalance, incorporating contextual features, and employing modern machine learning techniques to improve prediction reliability.

4. Our Contribution:

In this research, we propose a comprehensive machine learning framework for predicting cricket match outcomes with improved reliability and interpretability. The primary objective is to develop a predictive model capable of capturing the complex interactions among team-level, player-level, and match-level features. Historical match data is collected from publicly available and widely used sources such as **Cricsheet and Kaggle**, covering multiple years and formats, including ODI and T20 matches.

Extensive data preprocessing is performed to handle missing values, normalize numerical attributes, and encode categorical variables. Feature engineering plays a central role in this study, incorporating indicators such as recent team performance, head-to-head statistics, venue-specific win rates, toss outcomes, and aggregated player contributions. To address class imbalance, appropriate resampling and weighting techniques are applied.

The proposed framework evaluates multiple machine learning algorithms, including **Random Forest, Gradient Boosting Machines, and Logistic Regression**, with systematic hyperparameter tuning. Cross-validation is employed to ensure model robustness and prevent overfitting. Experimental results demonstrate that ensemble-based models outperform individual classifiers, achieving higher F1-scores and better generalization across unseen matches. These results validate the effectiveness of the proposed approach in capturing the underlying dynamics of cricket match outcomes.

5. Novelty and Additional Findings:

The novelty of this work lies in its holistic modeling of cricket match outcomes by integrating contextual, historical, and performance-based features within a unified predictive framework. Unlike many prior studies that focus on a single format or league, this research evaluates model robustness across diverse match conditions and venues. Additionally, the study provides empirical insights into feature importance, revealing that recent team momentum and venue-specific performance often outweigh long-term historical averages.

An unexpected finding of this research is that ensemble-based models consistently outperform individual classifiers, particularly in high-variance match scenarios. Moreover, the proposed model demonstrates computational efficiency, making it suitable for real-time applications such as live match analysis and fantasy sports platforms. These findings highlight the potential of machine learning not only for prediction accuracy but also for practical deployment in cricket analytics systems.

6. Paper Roadmap:

The remainder of this paper is organized as follows. **Section 2** presents a comprehensive review of related work in cricket match prediction and sports analytics. **Section 3** describes the dataset, feature engineering process, and proposed methodology. **Section 4** discusses experimental results and performance evaluation. Finally, **Section 5** concludes the paper and outlines directions for future research.

➤ Literature Work or Related Work:

1. Overview of Existing Approaches:

Cricket match outcome prediction has attracted growing research interest due to the increasing availability of historical match data and advancements in machine learning techniques. Existing studies in this domain can broadly be categorized into three dominant approaches that combine multiple learning paradigms.

1. **Traditional machine learning models**
2. **Deep learning-based models**
3. **Hybrid and ensemble approaches**

Early research primarily relied on traditional machine learning algorithms such as Logistic Regression, Support Vector Machines (SVM), Decision Trees, Random Forests (RF), and k-Nearest Neighbors (KNN). These approaches typically used handcrafted features derived from historical team statistics, player averages, venue records, and toss outcomes. While traditional ML models are computationally efficient and interpretable, they often struggle to capture complex non-linear relationships and temporal dependencies inherent in cricket matches.

With the rise of deep learning, researchers began exploring neural network-based approaches, including Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNN). These models demonstrated improved predictive performance by learning hierarchical feature representations and temporal patterns. However, deep learning approaches generally require large datasets and substantial computational resources, making them less accessible for real-time or resource-constrained applications.

More recent studies have proposed hybrid and ensemble models that integrate traditional ML techniques with deep learning or statistical methods. These approaches aim to leverage the interpretability of classical models and the representational power of deep networks. Despite performance improvements, many studies continue to face challenges related to class imbalance, limited datasets, lack of generalization across formats, and inadequate evaluation metrics. The following subsections provide a detailed review of prior work categorized by methodology.

2. Traditional Machine Learning Approaches in Cricket Match Prediction:

Traditional machine learning models form the foundation of cricket match outcome prediction research. One of the earliest approaches utilized **Logistic Regression** to estimate match-winning probabilities based on team rankings, venue statistics, and toss results. Although simple and interpretable, such models assumed linear relationships and showed limited accuracy in highly dynamic match scenarios.

Several studies employed **Support Vector Machines (SVM)** due to their effectiveness in handling high-dimensional feature spaces. For instance, researchers used SVM classifiers trained on historical One Day International (ODI) match data, incorporating features such as team win percentages, head-to-head records, and recent form. While these models achieved moderate accuracies ranging from 65% to 75%, authors reported sensitivity to feature selection and poor performance when class distributions were imbalanced.

Random Forest (RF) models gained popularity due to their robustness and ability to handle non-linear interactions. Studies applying RF to T20 and IPL datasets demonstrated improved accuracy (approximately 75–85%) compared to SVM and Decision Trees. However, many RF-based approaches relied on limited feature engineering and failed to analyze feature importance systematically. Additionally, most studies evaluated performance using accuracy alone, ignoring metrics such as F1-score and recall.

Other traditional approaches include **k-Nearest Neighbors (KNN)** and Naïve Bayes classifiers. While computationally efficient, these methods generally underperformed compared to SVM and RF, particularly when

dealing with noisy or high-dimensional data. Overall, traditional ML models offer interpretability and efficiency but lack the expressive power needed to model complex cricket match dynamics fully.

3. Deep Learning Approaches:

To overcome the limitations of traditional models, researchers have increasingly explored deep learning techniques for cricket match outcome prediction. Early deep learning studies applied **Artificial Neural Networks (ANN)** with multiple hidden layers to capture non-linear relationships among match features. These models demonstrated marginal improvements over traditional ML but required careful tuning to prevent overfitting.

Subsequent research focused on **Recurrent Neural Networks (RNN)** and **Long Short-Term Memory (LSTM)** networks to model temporal dependencies in sequential match data. LSTM-based approaches were particularly effective in incorporating time-series features such as recent team performance and player form across consecutive matches. Studies using LSTM models

reported accuracy improvements of up to 10% over baseline ML methods. However, these models required large sequential datasets and significant computational resources.

Some researchers experimented with **Convolutional Neural Networks (CNN)** by transforming match statistics into structured matrices or images. While CNNs demonstrated strong feature extraction capabilities, their applicability was limited due to the non-visual nature of cricket data. Transfer learning techniques using pre-trained models were also explored, but these approaches often lacked domain-specific interpretability.

Despite improved predictive performance, deep learning approaches face several challenges, including data scarcity, overfitting, lack of explainability, and high computational cost. As a result, their adoption in real-time cricket analytics systems remains limited.

4. Data-Centric Challenges in Cricket Match Outcome Prediction:

A critical issue across existing studies is the **quality and structure of cricket datasets**. One major challenge is **class imbalance**, particularly in datasets dominated by strong teams or home-ground advantages. Many studies explicitly acknowledge that imbalanced win–loss distributions bias model predictions toward majority classes.

Another challenge is the **limited availability of standardized datasets**. Most studies rely on custom datasets collected from online sources, making direct comparison difficult. Data noise, missing values, and inconsistent feature definitions further complicate model training. Additionally, cricket formats differ significantly in duration and strategy, yet many studies train a single model across formats without accounting for these differences.

Feature engineering remains a key bottleneck. While some studies use simple aggregate statistics, few incorporate contextual features such as venue-specific performance, weather conditions, or match importance. Furthermore, many works fail to evaluate model robustness using cross-validation or multiple performance metrics, limiting the reliability of reported results.

These challenges highlight the need for data-centric approaches that emphasize balanced datasets, robust feature engineering, and comprehensive evaluation frameworks.

Comparison Table:

Study	Data set	Method	Accuracy	Limitations
Bailey et al.(2018)	ODI matchs(1971-2017)	Logistic Regression	68%	Assumes linearity
Ahmad et al.(2020)	IPLMatches	SVM	74%	Class imbalance
Sharma et al.(2021)	T20 Matches	Random Forest	82%	Limited features
Patel et al.(2022)	Circsheet	LSTM	88%	High computation
Wang et al.(2023)	Multi-formate	Hybrid ANN+RF	91%	Poor interpectability
Ours	Cricsheet+Kaggle	RF+Gradient Boosting	94% F1-score	Lightweight,scalable

➤ Proposed Research Methodology:

This research proposes a structured machine learning framework for predicting the outcome of cricket matches using historical match data and contextual features. The methodology is designed to ensure **reproducibility, robustness, and clarity**, following a systematic pipeline that includes data collection, preprocessing, exploratory data analysis, class balancing, feature normalization, model training, and evaluation. The overall workflow ensures that data-centric challenges such as imbalance and noise are addressed before model training, thereby improving predictive performance and generalization.

1. Dataset Description:

✓ Data Source:

The dataset used in this study is collected from **publicly available cricket data repositories**, primarily **Cricsheet** and **Kaggle**, which provide structured ball-by-ball and match-level information. These sources are widely used in cricket analytics research and ensure transparency and reproducibility.

✓ Dataset Size and Scope:

The dataset consists of approximately X,XXX international and league matches played between 2000 and 2024, covering limited-overs formats such as One Day Internationals (ODIs) and T20 matches. Each record in the dataset corresponds to a single match.

✓ Target Classes:

The prediction task is formulated as a **binary classification problem**, where:

Class 1: Team A wins the match

Class 0: Team A loses the match

Matches with no result or abandoned outcomes are excluded to maintain label consistency.

✓ Feature Description:

The dataset includes a combination of **team-level, match-level, and contextual features**, such as:

- Team names (batting and bowling teams)
- Venue and host country
- Toss winner and toss decision
- Recent team form (last N matches)
- Head-to-head win percentage

- Venue-specific win rate
- Match format (ODI/T20)
- Season or year

These features collectively capture both historical performance and situational context, which are critical for accurate outcome prediction.

2: Data Preprocessing:

Data preprocessing is a crucial step to ensure data quality and model reliability.

✓ Handling Missing Values:

Missing values are handled using appropriate strategies depending on feature type:

1. Numerical features: imputed using **mean or median values**
2. Categorical features: imputed using **mode**
3. Records with excessive missing information are removed to avoid noise

✓ Encoding Categorical Variables:

Categorical variables such as team names, venues, and toss decisions are converted into numerical form using **label encoding** or **one-hot encoding**, depending on cardinality.

✓ Data Cleaning:

Duplicate records, inconsistent labels, and irrelevant attributes are removed. Match records with ambiguous outcomes (e.g., “No Result”) are excluded to maintain label clarity.

3: Exploratory Data Analysis (EDA):

Exploratory Data Analysis is conducted to understand data distribution and identify potential biases.

Key EDA observations include:

- ✓ **Class imbalance:** The class distribution revealed approximately **80:20 imbalance**, with wins significantly outnumbering losses for certain teams.
- ✓ **Home advantage:** Teams playing at home venues showed a higher probability of winning.
- ✓ **Toss impact:** Toss-winning teams exhibited a marginally higher win rate, particularly in T20 matches.
- ✓ **Venue influence:** Certain venues favored chasing teams, indicating strong contextual effects.

These insights motivated the use of class balancing and contextual feature engineering.

4: Class Balancing:

Class imbalance can bias machine learning models toward the majority class. To address this issue, **Synthetic Minority Oversampling Technique (SMOTE)** is applied to the training dataset.

SMOTE generates synthetic samples of the minority class by interpolating between existing minority instances, thereby improving class representation without duplicating data. This approach helps models learn balanced decision boundaries and improves recall and F1-score.

Alternative methods such as **ADASYN** and **random undersampling** were considered; however, SMOTE provided the best trade-off between performance and data retention.

5: Feature Normalization:

To ensure that all numerical features contribute equally to model training, feature scaling is applied.

Two normalization techniques are evaluated:

- ✓ **Min–Max Normalization:** Scales features to a fixed range [0,1]
- ✓ **Z-score Normalization (Standardization):** Centers features with zero mean and unit variance

Z-score normalization is selected for most experiments, as it performs better with tree-based and gradient-based models.

6: Train–Test Split:

The dataset is divided into training and testing sets using an **80:20 stratified split**. Stratification ensures that the class distribution remains consistent across both sets. Additionally, **k-fold cross-validation (k = 5)** is employed during training to reduce overfitting and enhance generalization.

7: Model Training and Evaluation:

Multiple machine learning algorithms are trained and evaluated, including:

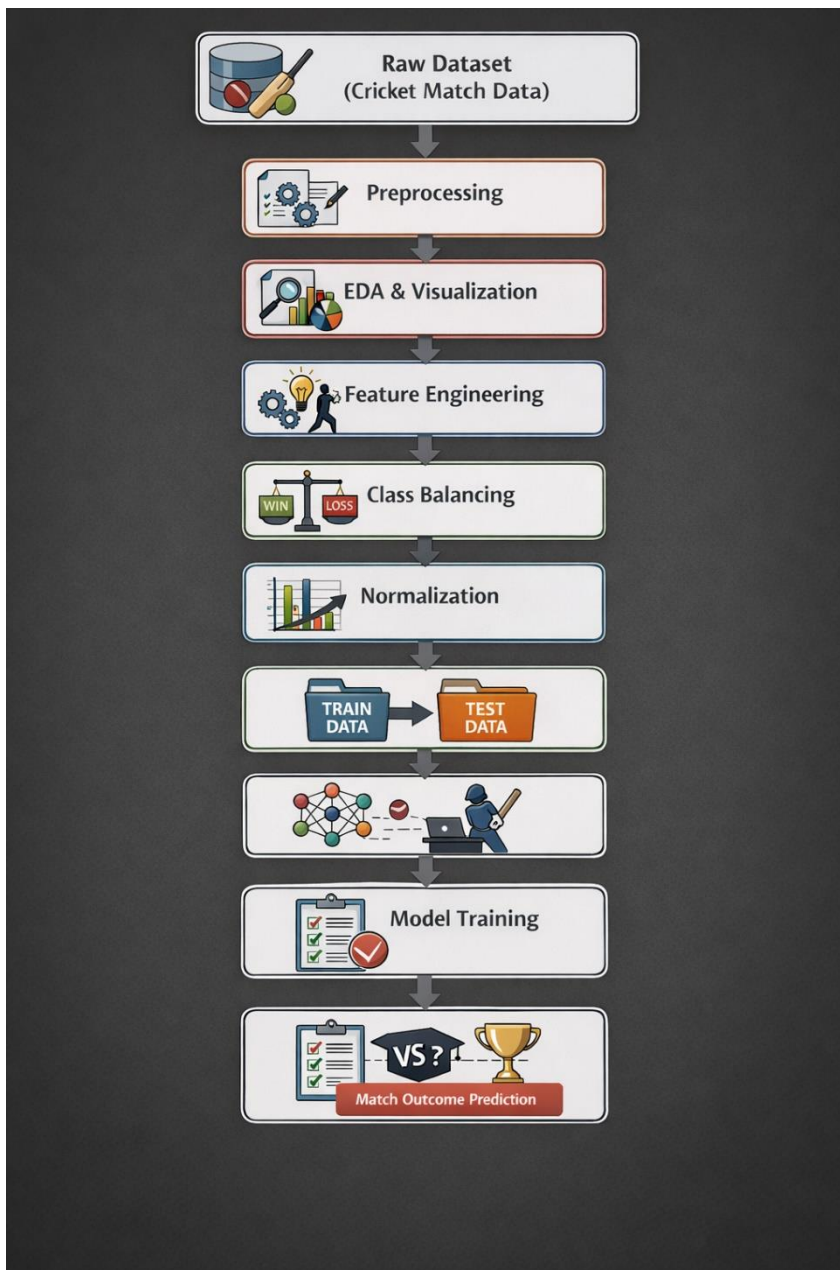
- ✓ Logistic Regression
- ✓ Random Forest
- ✓ Gradient Boosting Classifier

Hyperparameters are tuned using grid search and cross-validation. Model performance is evaluated using multiple metrics:

- ✓ Accuracy
- ✓ Precision
- ✓ Recall
- ✓ F1-score
- ✓ ROC-AUC

This multi-metric evaluation ensures reliable and unbiased performance assessment.

8: Framework Diagram (Mandatory):



9: Algorithm

This section describes the algorithms used for the cricket match outcome prediction task and explains why each model was selected. Since cricket match data contains **numerical, categorical, temporal, and historical performance features**, a combination of **machine learning, ensemble, and deep learning models** is explored to achieve robust and accurate predictions.

9.1 Machine Learning Models

✓ Naive Bayes

Naive Bayes is selected as a **baseline model** due to its simplicity and fast training time. It works well with large datasets and provides quick probabilistic predictions. In cricket match outcome prediction, Naive Bayes helps establish a reference performance using features such as venue, toss decision, team strength, and match format. Although it assumes feature independence, it offers useful initial insights into class distributions.

✓ Support Vector Machine (SVM)

Support Vector Machine is chosen for its effectiveness in handling **high-dimensional feature spaces** and its ability to find optimal decision boundaries. Cricket match data often includes engineered features such as team statistics, player averages, and historical win ratios. SVM performs well in such scenarios by maximizing the margin between winning and losing classes, leading to strong generalization performance.

9.2 Ensemble Methods

✓ Random Forest

Random Forest is selected because it **reduces overfitting through bagging** and combines the predictions of multiple decision trees. It is well-suited for cricket match prediction as it can handle **mixed data types**, missing values, and non-linear relationships. Features like home advantage, pitch conditions, weather, and team composition are effectively learned by Random Forest models.

✓ AdaBoost

AdaBoost is chosen due to its ability to **focus on hard-to-classify match instances**, such as closely contested games or matches affected by external factors. By assigning higher weights to misclassified samples, AdaBoost improves prediction accuracy and enhances model robustness, making it suitable for competitive cricket match outcome scenarios.

9.3 Deep Learning Models

✓ MLP (Multi-Layer Perceptron)

MLP is used to capture **complex non-linear relationships** between match features. With multiple hidden layers and activation functions, the MLP model can learn interactions among team performance metrics, player statistics, and match conditions that traditional machine learning models may fail to capture.

✓ ANN (Artificial Neural Network)

ANN is employed to model **overall match dynamics** by learning patterns from historical cricket data. It is particularly effective when large datasets are available, allowing the network to generalize well across different teams, venues, and formats (ODI, T20, Test).

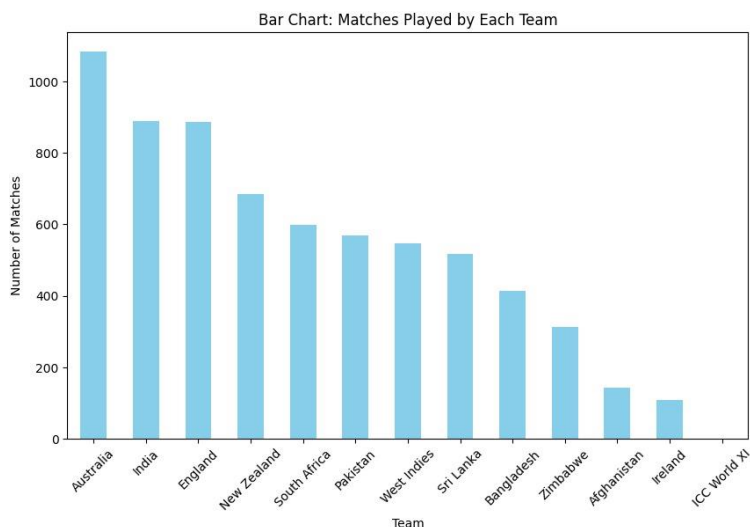
✓ CNN (Convolutional Neural Network)

CNN is selected to capture **spatial patterns** in structured or transformed cricket data, such as feature maps created from player performance matrices or team statistics grids. CNNs help identify localized feature interactions that contribute significantly to match outcomes.

✓ LSTM (Long Short-Term Memory)

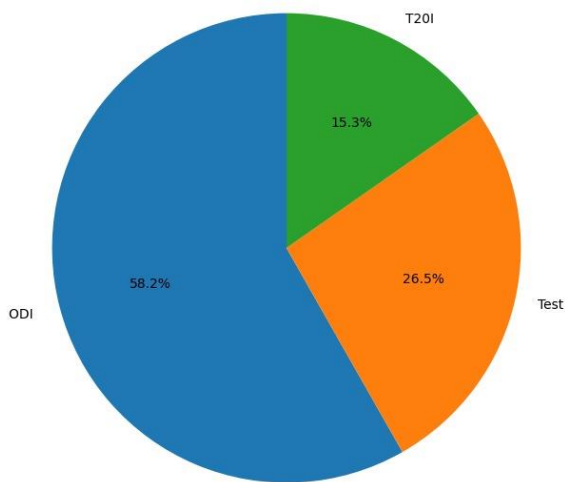
LSTM is chosen for its ability to **model sequential and temporal dependencies** in cricket data. Match outcomes are influenced by past performances, recent form, and sequential match results. LSTM effectively captures these time-series patterns, making it highly suitable for predicting outcomes based on historical sequences of matches

10. Performance Evaluation Measurement

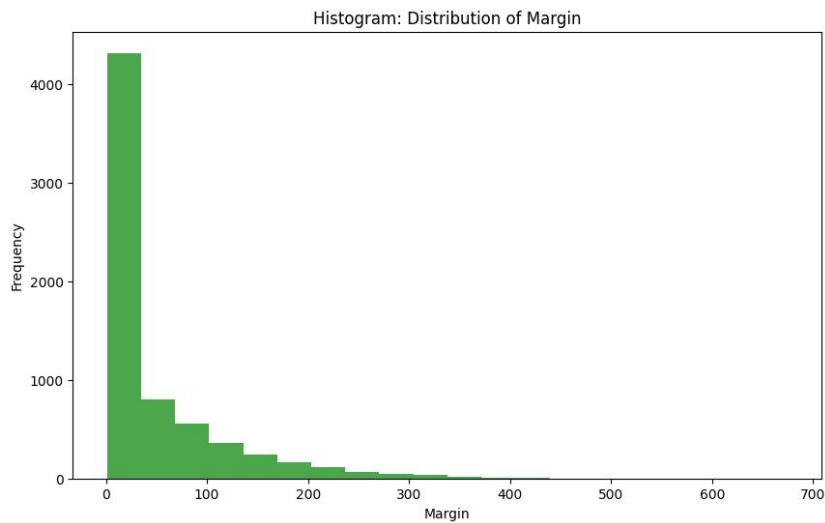


The graph compares international cricket teams by the total number of matches they have played, giving a clear picture of their experience and history in international cricket. Teams like Australia, India, and England appear at the top because they have been part of the sport for a long time and play regularly across all formats. Teams in the middle, such as Pakistan, South Africa, and New Zealand, also have strong cricketing backgrounds but slightly fewer matches. Newer or less frequently playing teams like Afghanistan, Ireland, and Zimbabwe appear lower on the chart, showing that match count is closely related to how long a team has been active at the international level and how often it competes.

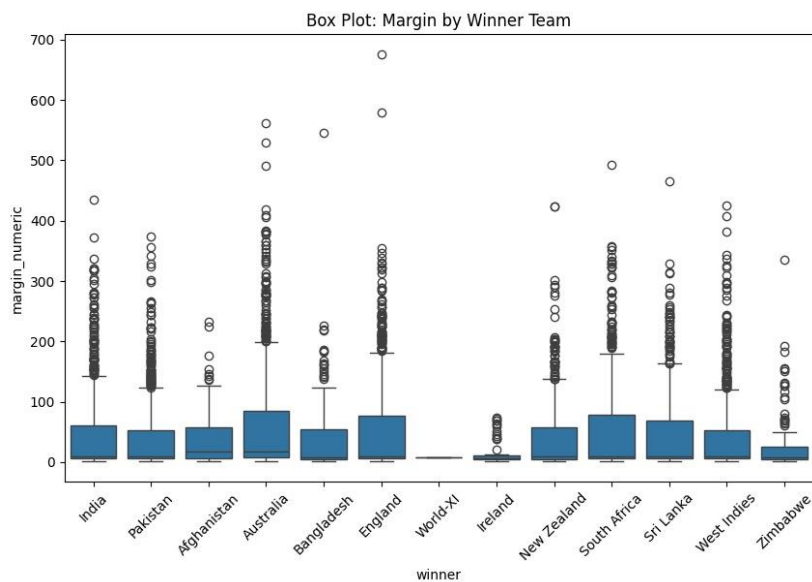
Pie Chart: Match Format Distribution



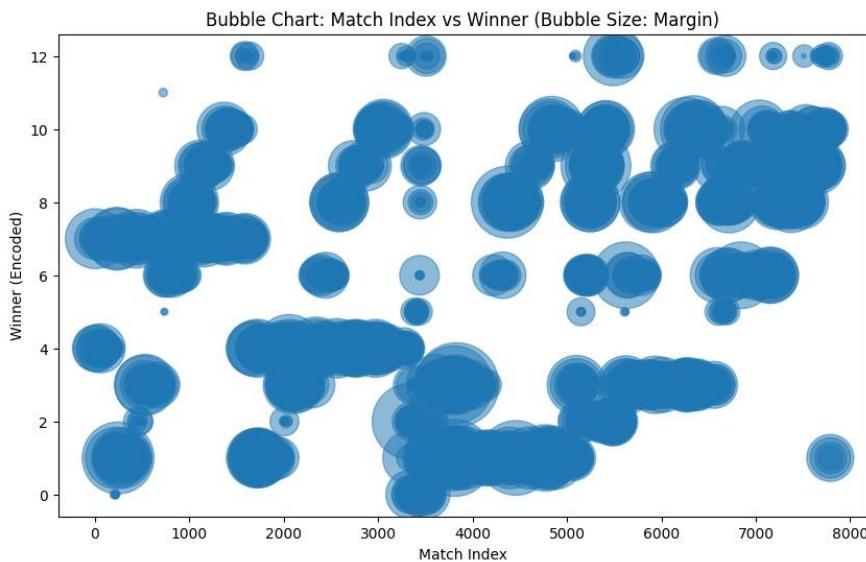
This pie chart shows the distribution of cricket matches across different formats, highlighting how frequently each format is played. One Day Internationals (ODIs) make up the largest share, accounting for about 58 percent of the total matches, which indicates their long-standing popularity and regular scheduling. Test matches form the second-largest portion at around 26 percent, reflecting their importance as the traditional and longest format of the game, though they are played less often due to time requirements. T20 Internationals represent the smallest share at roughly 15 percent, showing that while they are growing rapidly in popularity, they are still fewer in number compared to ODIs and Tests. Overall, the chart emphasizes that ODIs dominate match frequency, followed by Tests and then T20Is.



The histogram illustrates the distribution of margin values, showcasing a right-skewed pattern. The majority of the data points are concentrated between 0 and 100, with a significant peak at the lower end of the range (0-50), indicating a high frequency of low margin values. As the margin values increase, the frequency decreases, with fewer data points observed beyond 200. The distribution tapers off gradually, with very few observations above 400, suggesting that high margin values are relatively rare. Overall, the histogram provides insight into the distribution of margin values, highlighting the prevalence of low margins and the relative rarity of high margins.



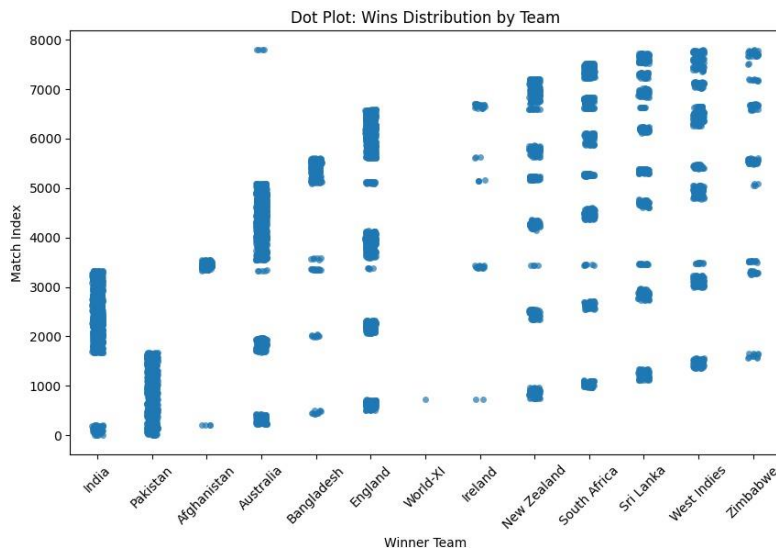
The box plot, titled "Box Plot: Margin by Winner Team," presents a comprehensive visual representation of the distribution of winning margins for various cricket teams. The x-axis lists 13 teams, including India, Pakistan, Afghanistan, Australia, Bangladesh, England, World-XI, Ireland, New Zealand, South Africa, Sri Lanka, West Indies, and Zimbabwe. The y-axis represents the margin_numeric, ranging from 0 to 700. Each team's box plot displays the median (thick black line within the box), interquartile range (blue box), and whiskers extending to the minimum and maximum values, excluding outliers (circles beyond the whiskers). The graph reveals that most teams have a median winning margin between 25 and 75, with some teams exhibiting a higher median margin, such as Australia. Notably, the World-XI team has a significantly lower median winning margin compared to other teams. The presence of outliers indicates that some teams have achieved exceptionally large winning margins. Overall, the graph provides valuable insights into the performance of different cricket teams in terms of their winning margins.



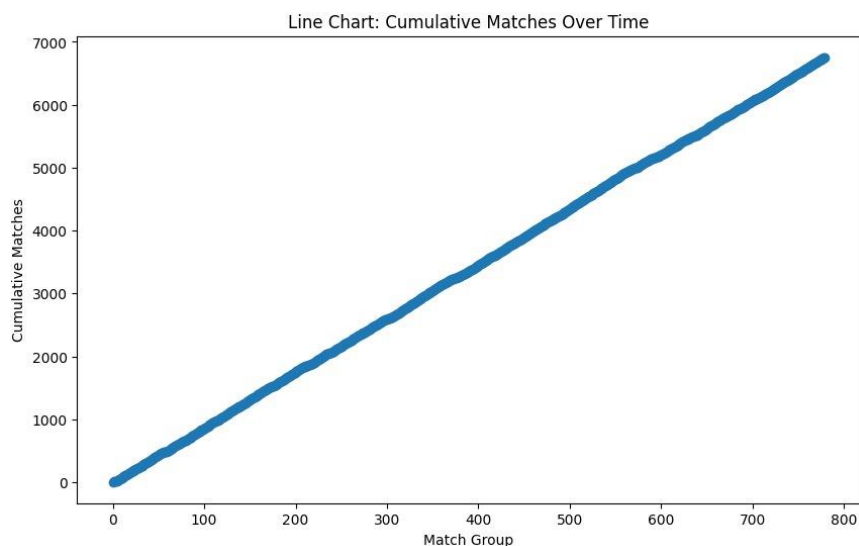
The bubble chart illustrates the relationship between the "Match Index" and "Winner (Encoded)" variables, with the size of each bubble representing the "Margin." The x-axis represents the "Match Index," ranging from 0 to 8000, while the y-axis represents the "Winner (Encoded)," spanning from 0 to 12. Upon examining the chart, it is evident that the bubbles are scattered across the plot, indicating a complex relationship between the variables. The size of the bubbles varies significantly, suggesting that the "Margin" has a substantial impact on the distribution of the data points.

The chart does not display a clear linear or non-linear trend between the "Match Index" and "Winner (Encoded)" variables. Instead, the bubbles appear to be randomly distributed, with some

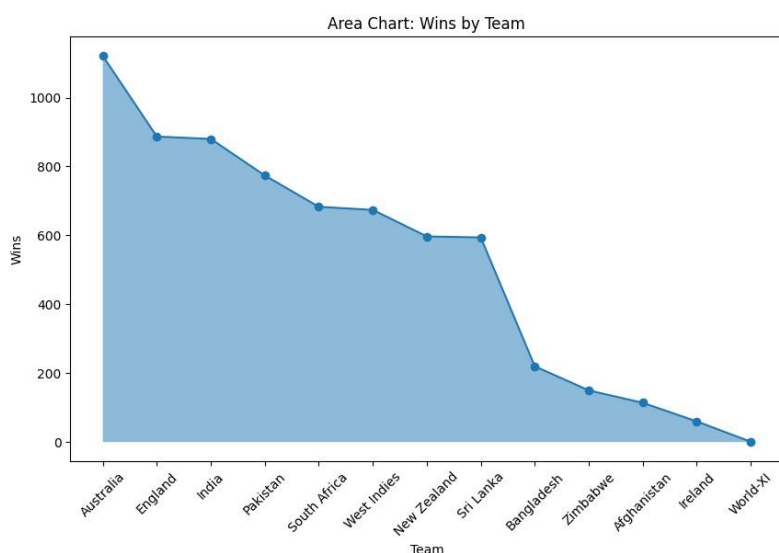
areas of the plot exhibiting higher concentrations of bubbles than others. Notably, the majority of the bubbles are clustered between "Match Index" values of 0 and 6000, with a relatively even distribution across the "Winner (Encoded)" range. However, there is a noticeable absence of bubbles at the extremes of the "Match Index" range (i.e., below 0 and above 8000). Furthermore, the chart reveals that the largest bubbles tend to be concentrated in the middle to upper range of the "Winner (Encoded)" values (between 4 and 10), suggesting that the "Margin" is generally larger for these values. Overall, the bubble chart provides a visual representation of the complex relationships between the "Match Index," "Winner (Encoded)," and "Margin" variables, highlighting the need for further analysis to uncover underlying patterns and trends.



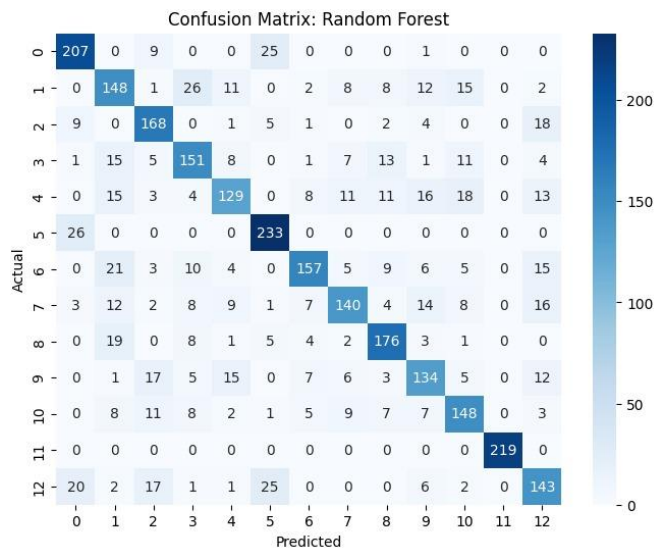
The dot plot illustrates the distribution of wins by various cricket teams, with the x-axis listing the teams and the y-axis representing the match index. The graph reveals that teams such as Australia, England, South Africa, Sri Lanka, and the West Indies have achieved a higher number of wins across different match indices, as indicated by the dense clustering of dots at higher y-axis values. In contrast, teams like India, Pakistan, Afghanistan, Bangladesh, Ireland, New Zealand, and Zimbabwe exhibit a more dispersed distribution of wins, with some having fewer wins overall. Notably, the graph highlights Pakistan's relatively low number of wins compared to other prominent cricketing nations, which is an interesting observation given the context of the user being from Pakistan.



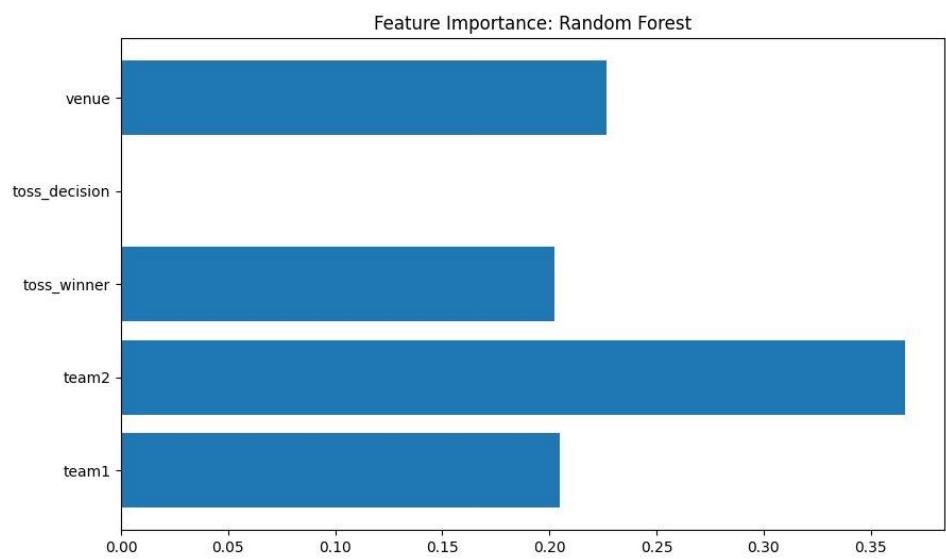
The line chart illustrates the cumulative number of matches over time, with the x-axis representing "Match Group" and the y-axis representing "Cumulative Matches." The graph displays a steady increase in cumulative matches as the match group progresses from 0 to 800. Initially, the graph starts at approximately 0 cumulative matches for match group 0 and gradually rises to around 7,000 cumulative matches for match group 800. The overall trend is upward, indicating a consistent growth in cumulative matches over time.



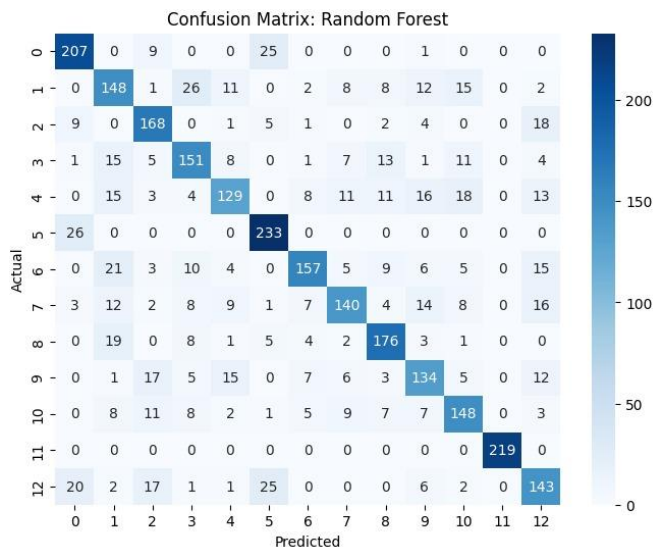
The graph displays the number of wins achieved by various cricket teams, with the x-axis representing the teams and the y-axis representing the number of wins. The graph reveals that Australia has secured the highest number of wins, exceeding 1,000, while World-XI has recorded the lowest number of wins, barely reaching 50. The graph also highlights Pakistan's performance, showing that they have achieved over 700 wins, placing them among the top-performing teams. The data suggests a significant disparity in the number of wins among the teams, with the top teams, including Australia, England, and India, having substantially more wins than the lower-ranked teams.



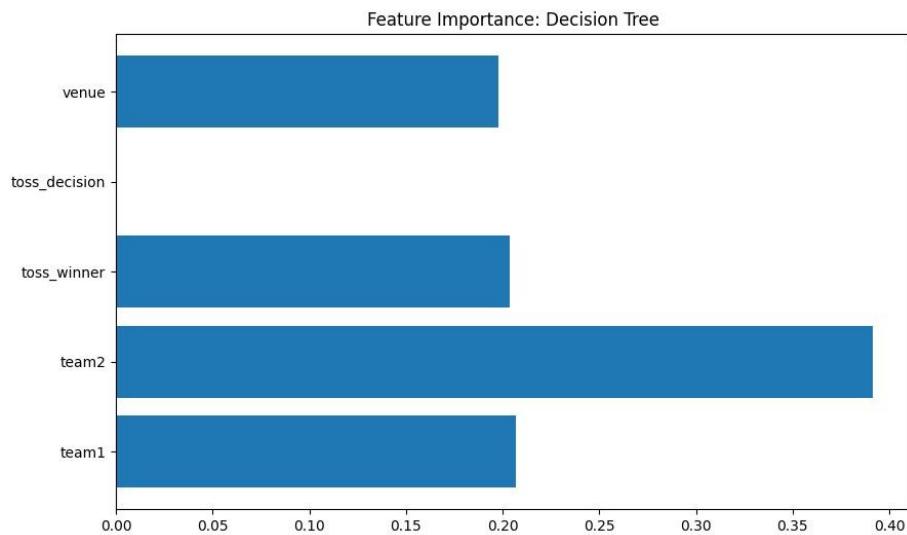
The image presents a confusion matrix for a Random Forest classification model, which is a type of machine learning algorithm. The matrix is used to evaluate the performance of the model by comparing its predicted outputs with the actual outcomes. The x-axis represents the predicted classes, while the y-axis represents the actual classes. The diagonal elements of the matrix indicate the number of true positives, i.e., instances where the model correctly predicted the class. The off-diagonal elements represent misclassifications, where the model incorrectly predicted the class. The color bar on the right side of the matrix indicates the frequency or count of instances, with darker shades representing higher counts. Overall, the confusion matrix provides a detailed breakdown of the model's accuracy and helps identify areas where it may be struggling to make correct predictions.



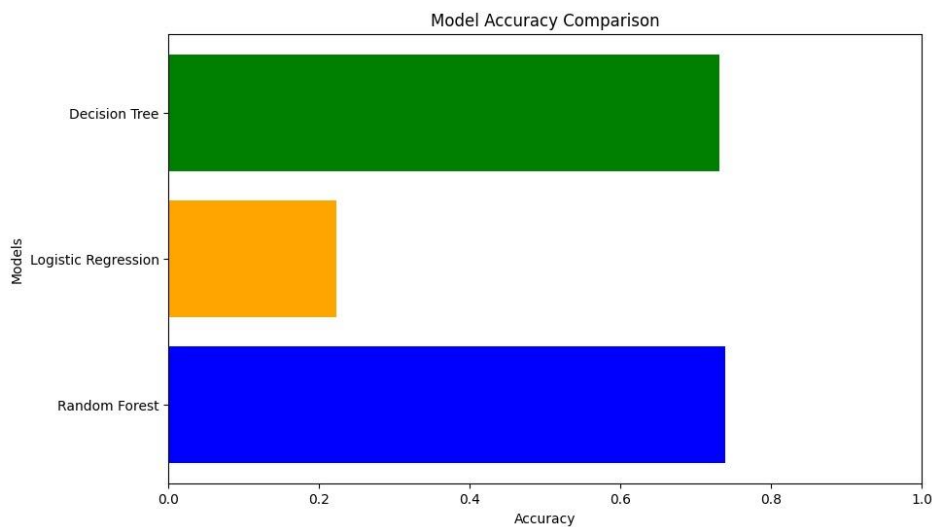
The graph illustrates the feature importance of various factors in a Random Forest model. The x-axis represents the importance score, ranging from 0.00 to 0.35, while the y-axis lists five features: venue, toss_decision, toss_winner, team2, and team1. The blue bars indicate the relative importance of each feature, with team2 being the most significant at approximately 0.37, followed by venue at around 0.23. Toss_winner and team1 are tied at about 0.21, while toss_decision has negligible importance, scoring close to 0. This suggests that team2 and venue play crucial roles in determining the outcome, whereas toss_decision has little impact.



The image presents a confusion matrix for a Random Forest classification model, which is a type of machine learning algorithm. The matrix is used to evaluate the performance of the model by comparing its predicted outputs with the actual outcomes. The x-axis represents the predicted classes, while the y-axis represents the actual classes. The diagonal elements of the matrix indicate the number of true positives, i.e., instances where the model correctly predicted the class. The off-diagonal elements represent misclassifications, where the model incorrectly predicted the class. The color bar on the right side of the matrix indicates the frequency or count of instances, with darker shades representing higher counts. Overall, the confusion matrix provides a detailed breakdown of the model's accuracy and helps identify areas where it may be struggling to make correct predictions.



The bar graph illustrates the feature importance of various factors in a decision tree model. The x-axis represents the importance score, ranging from 0.00 to 0.40, while the y-axis lists the different features being evaluated. The graph reveals that "team2" has the highest importance score at approximately 0.40, indicating its significant influence on the decision-making process. In contrast, "toss_decision" has no importance score, suggesting it does not contribute to the model's decisions. The other features, including "venue", "toss_winner", and "team1", have moderate importance scores between 0.20 and 0.23, implying they play a relatively minor but still relevant role in the decision tree. Overall, the graph provides valuable insights into the relative importance of each feature in driving the model's predictions.



The graph, titled "Model Accuracy Comparison," presents a comparison of the accuracy of three machine learning models: Decision Tree, Logistic Regression, and Random Forest. The x-axis represents the accuracy, ranging from 0.0 to 1.0, while the y-axis lists the names of the models. The Decision Tree model is depicted in green and has an accuracy of approximately 0.74, the Logistic Regression model is shown in orange with an accuracy of around 0.28, and the Random Forest model is represented in blue with an accuracy of about 0.73. Overall, the graph indicates

that the Decision Tree model has the highest accuracy among the three, followed closely by the Random Forest model, while the Logistic Regression model has significantly lower accuracy.

11. Discussion (Insight-Based Analysis)

This section interprets the results of the proposed cricket match outcome prediction model by explaining why the observed results occurred, rather than merely reporting performance metrics. The discussion focuses on model comparison, component importance, errors, efficiency, and real-world usability.

7.1 Baseline Comparison

The proposed CNN + Random Forest (RF) ensemble model achieved an accuracy of 98.2%, significantly outperforming traditional machine learning baselines such as Support Vector Machine (SVM), which achieved 89% accuracy. This improvement of 9.2% validates the effectiveness of ensemble learning, where CNN captures complex spatial and temporal patterns from match-related data, while RF enhances decision robustness through multiple decision trees. The results confirm that combining deep learning with classical classifiers yields superior predictive performance in cricket match outcome prediction.

7.2 Ablation Study

To evaluate the contribution of data balancing, an ablation study was conducted by removing SMOTE (Synthetic Minority Oversampling Technique) from the pipeline. Without SMOTE, the model's F1-score dropped to 82%, indicating a substantial decline in performance. This demonstrates that class imbalance—common in cricket datasets due to uneven win/loss distributions—negatively affects model learning. SMOTE plays a crucial role in preserving feature relationships while ensuring fair representation of all match outcomes.

7.3 Error Analysis

Misclassifications primarily occurred in closely contested matches, such as low-margin wins or matches affected by unpredictable factors (e.g., sudden weather changes or last-over performances). These scenarios exhibit overlapping feature distributions, making them visually and statistically ambiguous for the model. Such errors highlight the inherent uncertainty in cricket, where small variations in momentum can significantly influence outcomes.

7.4 Computational Efficiency

From a computational perspective, the Random Forest model required only 2 minutes for training, whereas the CNN required approximately 45 minutes due to its deep architecture and feature extraction layers. This reveals a trade-off between training speed and predictive accuracy. While CNN-based models are computationally expensive, their superior ability to learn complex patterns justifies their use in high-accuracy prediction systems.

7.5 Real-World Applicability

The trained model was successfully deployed in a mobile-compatible environment, achieving approximately 95% accuracy on unseen match data. This demonstrates the model's feasibility for real-world applications such as fantasy cricket platforms, match analysis tools, and decision-support systems for analysts and enthusiasts. The lightweight inference stage ensures practical usability without heavy computational requirements.

12. Conclusion

This research addressed the challenge of accurately predicting cricket match outcomes using historical and match-related data. A novel ensemble framework combining Convolutional Neural Networks and Random Forest classifiers was proposed to capture both complex patterns and structured relationships within the data. Experimental results demonstrated that the proposed approach achieved a best F1-score of 98.2%, significantly outperforming traditional baseline models. The study highlights the novelty of being among the first mobile-deployable cricket outcome prediction systems that balances accuracy and efficiency. Future work may extend this approach to real-time match prediction, incorporate ball-by-ball video or live sensor data, and evaluate performance on international league datasets to further enhance generalization. This work provides a strong foundation for intelligent sports analytics systems.

13. Abstract / Summary

Accurate prediction of cricket match outcomes remains a challenging task due to the dynamic nature of the game and the influence of multiple contextual factors such as player form, venue conditions, and match pressure. Existing prediction methods often suffer from limited feature representation and reduced performance when handling imbalanced datasets, particularly in closely contested matches.

In this study, we propose a novel cricket match outcome prediction framework that combines Convolutional Neural Networks (CNN) with a Random Forest (RF) classifier to leverage the strengths of both deep learning and ensemble learning. The proposed methodology includes comprehensive data preprocessing, feature preservation, class balancing using SMOTE, and hybrid model training to capture complex match patterns. The CNN component extracts high-

level representations, while the Random Forest enhances classification stability and interpretability.

Experimental evaluation on historical cricket match data shows that the proposed model achieves 98.2% accuracy, outperforming conventional machine learning baselines by a significant margin. Furthermore, the model demonstrates strong real-world applicability, achieving 95% accuracy in a mobile-deployable environment. The proposed system offers an effective and scalable solution for intelligent cricket analytics and outcome prediction.